



UNIVERSITY OF
LIVERPOOL

Application Development with C++ (ELEC362)

Lecture 13: Basics of Qt

mihasan@liverpool.ac.uk

Previous lecture

- The basics of software engineering were introduced and discussed.
- The waterfall model was discussed with its advantages and disadvantages.
- The requirement analysis, software design, validation and verification, and maintenance were discussed.
- The incremental model was presented and discussed.

This lecture

- What is covered in this lecture?

1. Introduction to Qt library. 2. How to deploy a stand-alone app using Qt.

- Why it is covered?

To understand how Qt is structured as a library and how an application can be made independent from Qt.

- How are topics covered in this lecture:

Live demonstration

Why Qt??

- There are many C++ libraries aimed at GUI application development, including GTK+, Qt, MFC, and JUCE.
- Qt is one of the most widely supported GUI libraries, with a lot of online documentation, examples, tutorials, and supportive community.
- It can be integrated very well with other libraries such as OpenGL (graphics library) and Boost (math-based libraries).
- Qt is a cross-platform IDE that creates applications working on Windows, Linux, Mac-OS, Android, and many embedded systems.

Why Qt??

- Examples of known software built using Qt:



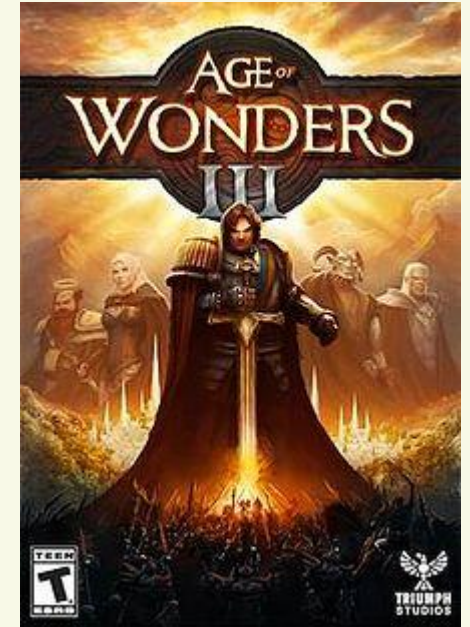
- Examples of known companies that use Qt for their software development:



Mercedes-Benz



PEUGEOT



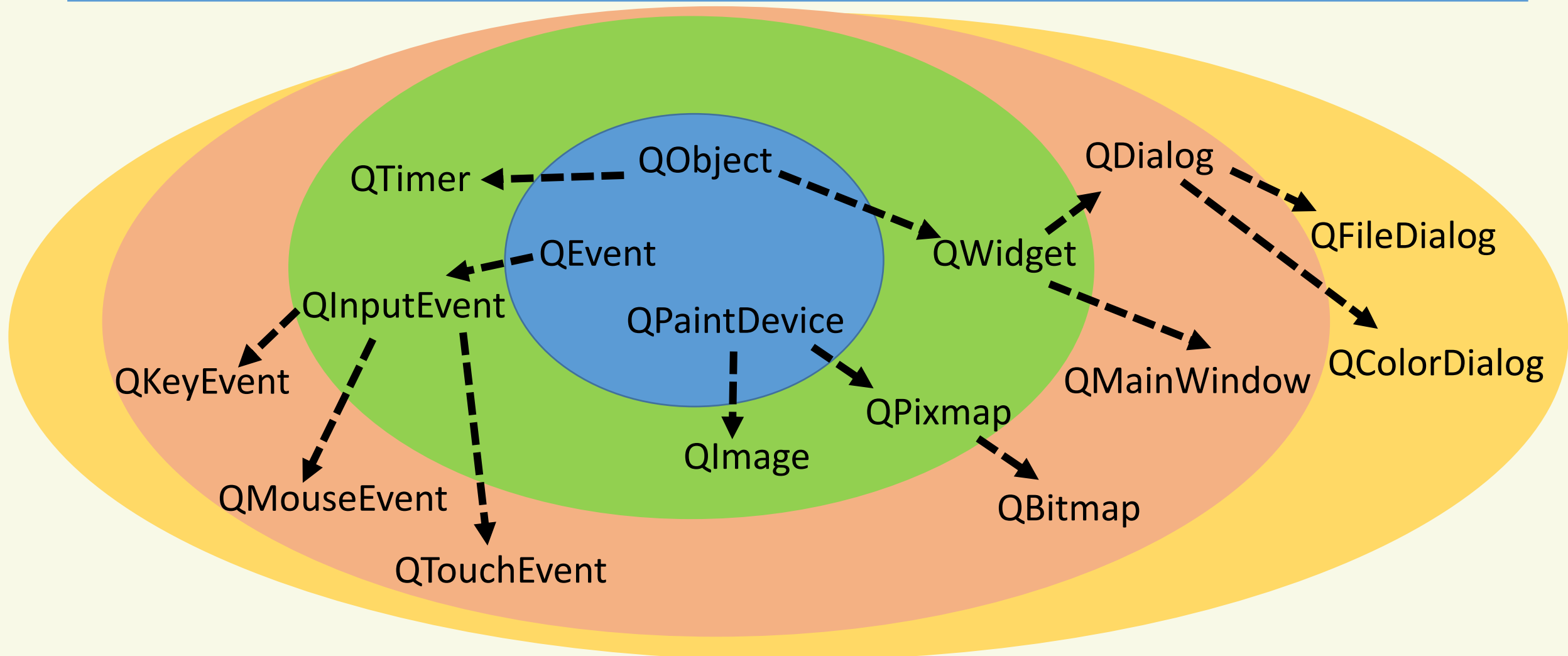
Qt resources and learning outcomes

- Books: Learn Qt 5: N Sherriff (2018).
C++ GUI Programming with Qt, J Blanchette & M Summerfield (2008).
- Online documentation: <https://doc.qt.io/>
<https://wiki.qt.io/Main>
- Online developers community: <https://forum.qt.io/>
- What you will learn is:
 1. How to use Qt to develop Graphic User Interface (GUI)-based applications.
 2. How to effectively use online documentations and resources for self-learning in programming context.

Qt basics

- Qt has been created in 1991, since then it has been growing.
- Qt is a C++ Library used to develop GUI applications for cross-platform applications.
- It is available under a commercial license and in a open-source licenses.
- It consists of more than 2100 classes!! Each class has more than 7-10 functions!!
- Full list of classes can be found here: <https://doc.qt.io/qt-6/classes.html>
- The name of every class in Qt starts with a “Q”.
- The class names are carefully chosen to reflect their purpose.

Qt class hierarchy



- QUESTION: Do you notice anything about the names of the classes ??

Qt and C++ relation

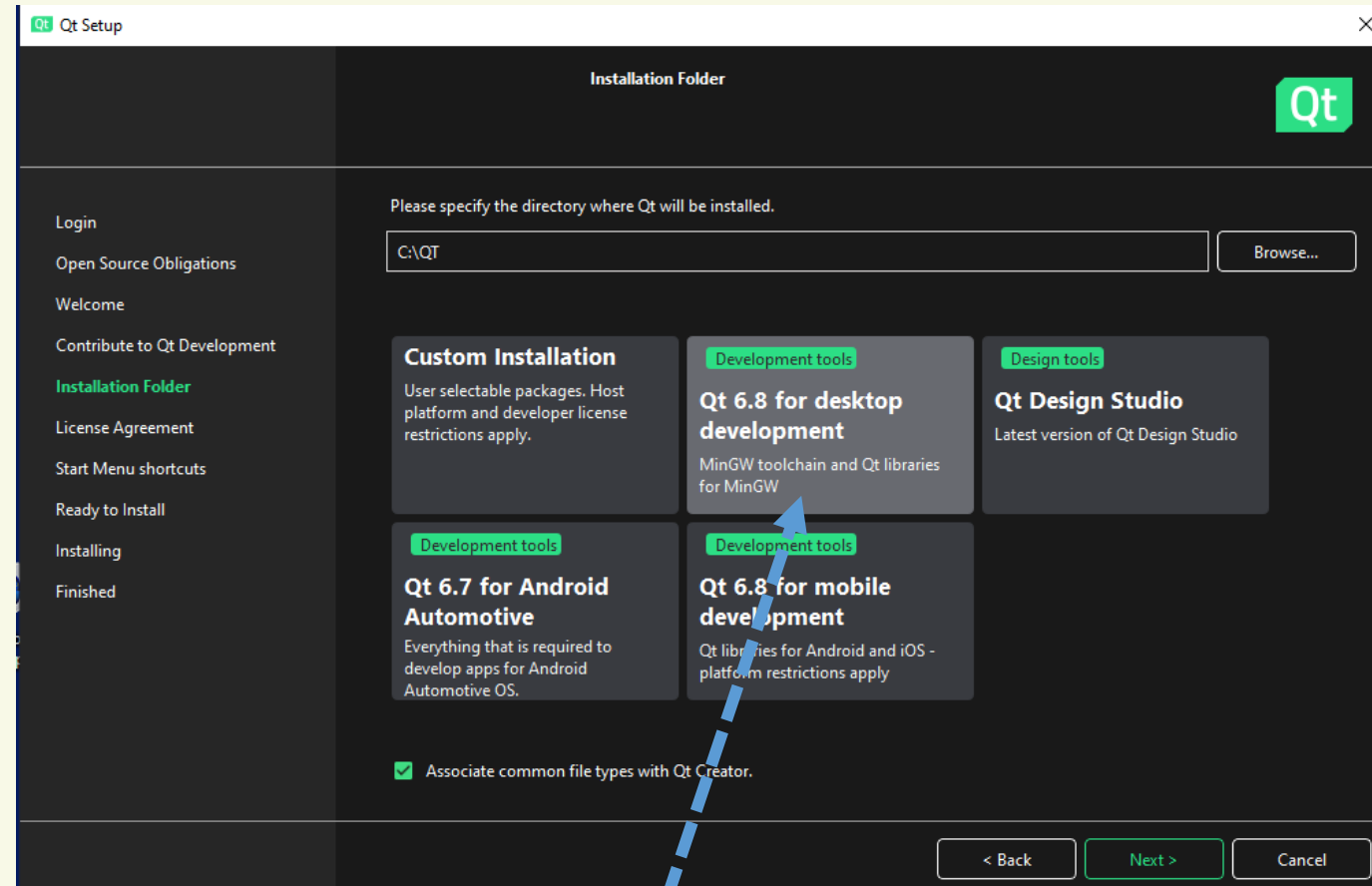
- Qt was developed using C++.
- Qt follows C++ syntax rules.
- Many of the widely used classes/containers in C++ were replaced in Qt:

C++ name	Qt equivalent
string	QString
vector	QVector
list	QList

- To make the use of Qt more efficient, a new language called “QML” was developed.

Installation of Qt

- In the labs and at the library, Qt is already pre-installed.
- For downloading and installing Qt on your computer/laptop, go to:
<https://www.qt.io/download-open-source>
- Make sure that the kit installed is Qt 6.9.



Choose a package that matches the platform of the application

Qt creator

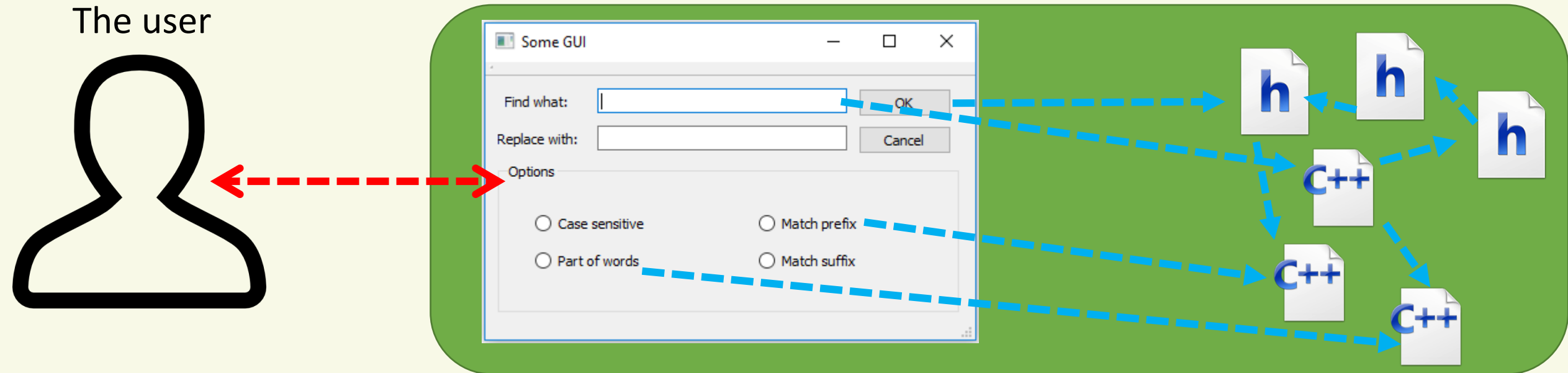
- The name of IDE software for Qt is “Qt creator”.
- The name of the GUI editing part of Qt creator is known as “Qt designer”.
- A Qt creator walkthrough (demonstration).
- Considering that 2100 classes is a large number to handle, classes are categorised into “modules”, each module is particularly suited for the hardware of the platform it is developed for.
- A full list of modules is given here: <https://doc.qt.io/qt-6/qtmodules.html>

How do GUI applications work?

- GUI applications work as follows:

The user

The application



What the application developers do is: 1. Design the GUI using objects from the library.

2. Connect the objects to one another in a functional way.

Deploying an application

- Qt creates the application easily within the IDE.
- The IDE generates an executable file that has dependencies. To make it a stand-alone application, the dependencies need to be put in the same folder as the exe file. Converting the application into a stand-alone app is known as “deploying”.
- For Windows application, one can use “Dependencies Walker” to determine which libraries an executable file need to run
(<http://www.dependencywalker.com/>)
- Deploying stand-alone executable from Qt (demonstration).

Deploying an application

- The needed libraries can be found in the “bin” folder in the installation folder of the Qt compiler installed for the platform (in our case the MinGW).
- The required libraries also depend on the type of build:

Libraries needed for windows build (all have .dll extension)

libgcc_s_seh-1, libstdc++-6, libwinpthread-1, Qt6Core, Qt6Gui, Qt6Widgets

- Note that the list of libraries may change based on the module used and the platform. Best use Dependency Walker to check.

Summary

- Justification for learning Qt was given.
- Basic structure of Qt was explained.
- An introduction to Qt Creator was given.
- Deploying an application using Qt was explained.